



Compiler Construction

Lab 03

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Task 1:

Design DFA for strings that contain at least one 0.

```
#include<iostream>

using namespace std;

int main() {
    string input;
    cout<<"Enter input: \n";
    cin >> input;

    int state = 0;

    for (char c : input) {
        if (state == 0) {
            if (c == '0') state = 1;
        }

    }

    if (state == 1)
        cout << "State Accepted";
    else
        cout << "State Rejected";

    return 0;
}
```

Output:

```
Enter input:
00001111
State Accepted
-----
Process exited after 5.536 seconds with return value 0
Press any key to continue . . .
```

```
Enter input:
1111
State Rejected
-----
Process exited after 3.591 seconds with return value 0
Press any key to continue . . .
```

Task 2:

Design DFA for strings that start with 1 and end with 0

```
#include <iostream>

#include <string>

using namespace std;

int main()
{
    string input;
    cout << "Enter input starting with 1 and end with 0: ";
    cin >> input;
    if (input[0]=='1' && input[input.length()-1]=='0')
    {

        cout << "String Accepted" << endl;
```

```
}  
else  
{  
    cout << "String Rejected" << endl;  
}  
  
return 0;  
}
```

Output:

```
Enter input starting with 1 and end with 0: 00000111  
String Rejected  
-----  
Process exited after 5.099 seconds with return value 0  
Press any key to continue . . .
```

```
Enter input starting with 1 and end with 0: 1110  
String Accepted  
-----  
Process exited after 4.725 seconds with return value 0  
Press any key to continue . . .
```